

EE1060: Differential Equations and Transform Techniques

Practice Problem Set - 01

Practice problems in topics covered till: Lecture 04

Note : The practice problem sets are designed for your personal review and are not required for submission or evaluation. These problems aim to reinforce your understanding of the material and help you prepare effectively. Some of the practice problems are sourced from the course references, and it is strongly recommended that you consult these references for additional practice problems. Due credit is also given to the TAs of the course, who have contributed to the creation of this problem set.

1. Determine whether the following differential equations are linear and identify their order.

(a) $x^2 \frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = 10x^3$

Solution: Second-order linear differential equation.

(b) $x^2 y \frac{dy}{dx} - 10x^3 = 0$

Solution: First-order nonlinear differential equation (due to the term $y \frac{dy}{dx}$).

(c) $x^2 \left(\frac{d^2 y}{dx^2} \right)^2 + \frac{dy}{dx} + y = 0$

Solution: Second-order nonlinear differential equation (due to the square of the second derivative term).

(d) $x^2 \left(\frac{d^2 y}{dx^2} \right)^2 + (3x + 4) \frac{dy}{dx} + y^2 = 0$

Solution: Second-order nonlinear differential equation (due to the y^2 term and the square of the second derivative term).

(e) $\sin x \frac{d^2 y}{dx^2} + \frac{dy}{dx} - xy = 0$

Solution: Second-order linear differential equation.

(f) $\frac{d^2 y}{dx^2} - \beta(1 - y^2) \frac{dy}{dx} + y = 0$

Solution: Second-order nonlinear differential equation (due to the y^2 term inside the derivative).

(g) $\frac{\partial y}{\partial t} + y \frac{\partial y}{\partial x} - \alpha \frac{\partial^2 y}{\partial x^2} = 0$, where α and β are constants.

Solution: First-order nonlinear partial differential equation (PDE) due to the $y \frac{\partial y}{\partial x}$ term.

(h) $\frac{d^2 y}{dx^2} + \alpha \frac{dy}{dx} + \beta y + ay^3 = A \cos \omega x.$

Solution: Second-order nonlinear differential equation (due to the y^3 term).

(i) $\frac{dy}{dx} - A(x) - B(x)y - c(x)y^2 = 0.$

Solution: First-order nonlinear differential equation (due to the y^2 term).

2. Show that $y = cx - x^2$ is a solution to the ODE. Also find the particular solution for the initial condition $y(1) = 2$.

$$x \frac{dy}{dx} - y + x^2 = 0 \quad (1)$$

Solution: To begin with, we will differentiate $y = \frac{1}{x+c}$. This results in

$$\frac{dy}{dx} = c - 2x \quad (2)$$

Substituting y' and y in the right hand side of the given differential equation results in

$$x \frac{dy}{dx} - y + x^2 = x(c - 2x) - (cx - x^2) + x^2 = 0 \quad (3)$$

This $y = cx - x^2$ is the solution to the given differential equation.

The value of the arbitrary constant for the given initial condition $y(1) = 2$ is

$$y(1) = 2 \implies c(1) - 1^2 = 2 \implies c = 3 \quad (4)$$

Thus, the particular solution for the given initial condition is $y = 3x - x^2$.

3. Show that $y = c_1 e^{-at} + c_2 e^{-bt}$ forms the 2-parameter solution to the differential equation

$$\frac{d^2 y}{dx^2} + (a+b) \frac{dy}{dx} + aby = 0 \quad (5)$$

Solution: The first and second derivatives of y are

$$\frac{dy}{dx} = -ac_1e^{-at} - bc_2e^{-bt} \text{ and } \frac{d^2y}{dx^2} = a^2c_1e^{-at} + b^2c_2e^{-bt}. \quad (6)$$

Substituting y , $\frac{dy}{dx}$, and $\frac{d^2y}{dx^2}$ into the given differential equation results in

$$a^2c_1e^{-at} + b^2c_2e^{-bt} + (a+b)(-ac_1e^{-at} - bc_2e^{-bt}) + ab(c_1e^{-at} + c_2e^{-bt}) \quad (7)$$

Simplification of which results in

$$\cancel{(a^2 - a(a+b) + ab)c_1e^{-at}} + \cancel{(b^2 - b(a+b) + ab)c_2e^{-bt}} = 0. \quad (8)$$

Thus, the given function is a solution to the given differential equation

4. Compute the current response of a series LC circuit with the initial condition $v_c(0) = V_0$, where v_c is the voltage across the capacitor. For reference, the current response of a series LC circuit is governed by the DE

$$\frac{d^2i}{dt^2} + \frac{1}{LC}i(t) = 0 \quad (9)$$

Solution: The 2-parameter solution to this differential equation, which involves two arbitrary constants, is

$$i(t) = A \cos(\omega t + \phi) \quad (10)$$

where A and ϕ are arbitrary constants and $\omega = \frac{1}{\sqrt{LC}}$.

For the given initial conditions, the constants A and ϕ in the general solution can be determined as follows.

- From the initial condition $i(0) = 0$, we have

$$i(0) = A \cos(\omega 0 + \phi) = 0 \implies A \cos \phi = 0 \implies \phi = \frac{\pi}{2} \quad (11)$$

- From the initial voltage condition $v_c(0) = V_0$, where $v_c(t) = \frac{1}{C} \int i(t)$, we get

$$v_c(0) = \underbrace{\frac{A}{\omega C} \sin(\omega t + \phi)}_{v_c(t)} = 0 \implies \frac{A}{\omega C} \sin \phi = V_0 \implies A = \omega C V_0 \quad (12)$$

Therefore, the particular solution corresponding to the specified initial conditions is $i(t) = \omega C V_0 \sin \omega t$.

5. Determine the value of α for which the current $i(t) = Ae^{-\alpha t} \cos(\omega t + \phi)$ where $\omega = \sqrt{\frac{1}{LC} - \alpha^2}$, forms the general solution to the second-order linear differential equation

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i = 0, \quad (13)$$

Solution: The first and second derivatives of $i(t) = Ae^{-\alpha t} \cos(\omega t + \phi)$ are

$$\frac{di}{dt} = Ae^{-\alpha t} (-\alpha \cos(\omega t + \phi) - \omega \sin(\omega t + \phi)) \text{ and } \frac{d^2 i}{dt^2} = Ae^{-\alpha t} ((\alpha^2 + \omega^2) \cos(\omega t + \phi) + 2\alpha\omega \sin(\omega t + \phi)) \quad (14)$$

Substituting the expressions for $\frac{di}{dt}$ and $\frac{d^2 i}{dt^2}$ into the given differential equation results in

$$LAe^{-\alpha t} ((\alpha^2 + \omega^2) \cos(\omega t + \phi) + 2\alpha\omega \sin(\omega t + \phi)) + RAe^{-\alpha t} (-\alpha \cos(\omega t + \phi) - \omega \sin(\omega t + \phi)) + \frac{1}{C} Ae^{-\alpha t} \cos(\omega t + \phi) = 0 \implies$$

$$Ae^{-\alpha t} \left[L((\alpha^2 + \omega^2) \cos(\omega t + \phi) + 2\alpha\omega \sin(\omega t + \phi)) + R(-\alpha \cos(\omega t + \phi) - \omega \sin(\omega t + \phi)) + \frac{1}{C} \cos(\omega t + \phi) \right] = 0. \quad (15)$$

This must hold for all t , so the coefficients of $\cos(\omega t + \phi)$ and $\sin(\omega t + \phi)$ must both be zero i.e.,

$$L(\alpha^2 + \omega^2) - R\alpha + \frac{1}{C} = 0 \text{ and } \omega(2L\alpha - R) = 0 \quad (16)$$

which results in

$$\alpha = \frac{R}{2L} \quad (17)$$

6. Determine the value of β for which $y(x) = e^{\beta x}$ is a solution to the differential equation

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0 \quad (18)$$

Solution: The first and second derivatives of $y = e^{\beta x}$ are

$$y'(x) = \beta e^{\beta x} \text{ and } y''(x) = \beta^2 e^{\beta x} \quad (19)$$

Substituting $y(x)$, $y'(x)$ and $y''(x)$ into the differential equation gives

$$(\beta^2 - 2\beta + 1)e^{\beta x} = 0 \quad (20)$$

For this to hold true for all x , we require $(\beta^2 - 2\beta + 1) = 0$, which gives $\beta = 1$.

7. Determine the value of n for which $y(x) = 2(x^n - 1) + ce^{-(x^n)}$ is a solution to the differential equation

$$\frac{dy}{dx} + 2xy = 4x^3 \quad (21)$$

Solution: Differentiating $y(x) = 2(x^n - 1) + ce^{-x^n}$ results in

$$y'(x) = 2nx^{n-1} - c(nx^{n-1})e^{-x^n}. \quad (22)$$

Substituting $y(x)$ and $y'(x)$ into the differential equation gives

$$y' + 2xy = 2nx^{n-1} - c(nx^{n-1})e^{-x^n} + 2x(2(x^n - 1) + ce^{-x^n}) = 4x^{n+1} + (2 - ce^{-x^n})nx^{n-1} - 4x + 2cxe^{-x^n} \quad (23)$$

For $n = 2$, the above term would simplify to $4x^3$.

8. Solve the first-order ODE

$$\frac{dy}{dx} = y^2 e^x \quad (24)$$

Solution: Rearrange the equation gives

$$\frac{1}{y^2} dy = e^x dx. \quad (25)$$

Integrating on both sides results in

$$\int \frac{1}{y^2} dy = \int e^x dx \implies -\frac{1}{y} = e^x + c \implies y = -\frac{1}{e^x + c} \quad (26)$$

where c is an arbitrary constant.

9. Compute the particular solution for the first-order ODE

$$(y^2 + 1)\frac{dy}{dx} = x \text{ with the initial condition } y(0) = 1 \quad (27)$$

Solution: Rearrange the equation gives

$$(y^2 + 1)dy = xdx. \quad (28)$$

Integrating on both sides results in

$$\int (y^2 + 1)dy = \int xdx \implies \frac{y^3}{3} + y = \frac{x^2}{2} + c \quad (29)$$

where c is an arbitrary constant. The arbitrary constant c can be computed using the given initial condition $y(0) = 1$ i.e.,

$$y(0) = 1 \implies \frac{1^3}{3} + 1 = 0 + C \implies C = \frac{4}{3} \quad (30)$$

Thus, the particular solution is

$$\frac{y^3}{3} + y = \frac{x^2}{2} + \frac{4}{3} \quad (31)$$